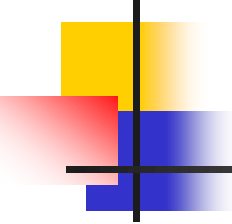
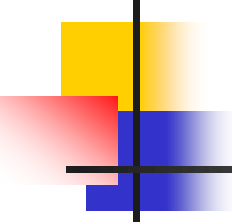


Lecture 3.0: Signaling



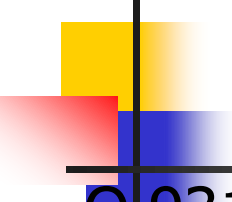
Introduction

- Signaling is used between the user and the network or between two network elements to exchange various control information like:
 - Traffic Descriptors
 - Service Descriptors
 - Channel Identifiers
- In other words, Signaling is used to dynamically establish, monitor, and release *connections* (including physical, virtual and logical connections)



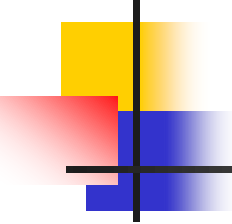
Examples

- Connection setup request
 - Off-hook signal from telephone to switch
- Connection setup acknowledge = Dial tone
- Destination address = Pulse or tone dialing
- Destination busy = Busy tone
- Destination Available = Ringing tone



Signaling Standards

- Q.931 = Basic Call Control for ISDN
- Q.932 = Extends/uses Q.931 for supplementary services (call forwarding, etc)
- Q.933 = Q.931 Extension/subset for Frame-relay
- Signaling inside the network is more sophisticated than that between the network and the subscriber
- Digital Subscriber Signaling System 1 (DSS1)
 - Call control signaling over the D channel
 - Q.931 + Q.932 + lower layers
- Signaling System 7 (SS7) deals with inside the network while DSS1 deals with outside.
- Q.2931 = Q.93B = Basic Call Control for B-ISDN



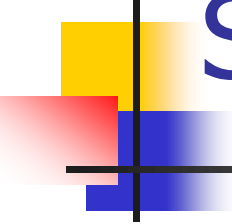
Signaling in Telecom Network

- Traditionally, the use of signaling in Telecom Networks was the bare minimum.
- It was restricted to establish/release a voice channel to allow telephonic conversation.
- For e.g., SS7 which has an advanced network architecture, provides feature-rich signaling

Signaling in

Virtual-Circuit-Based Network

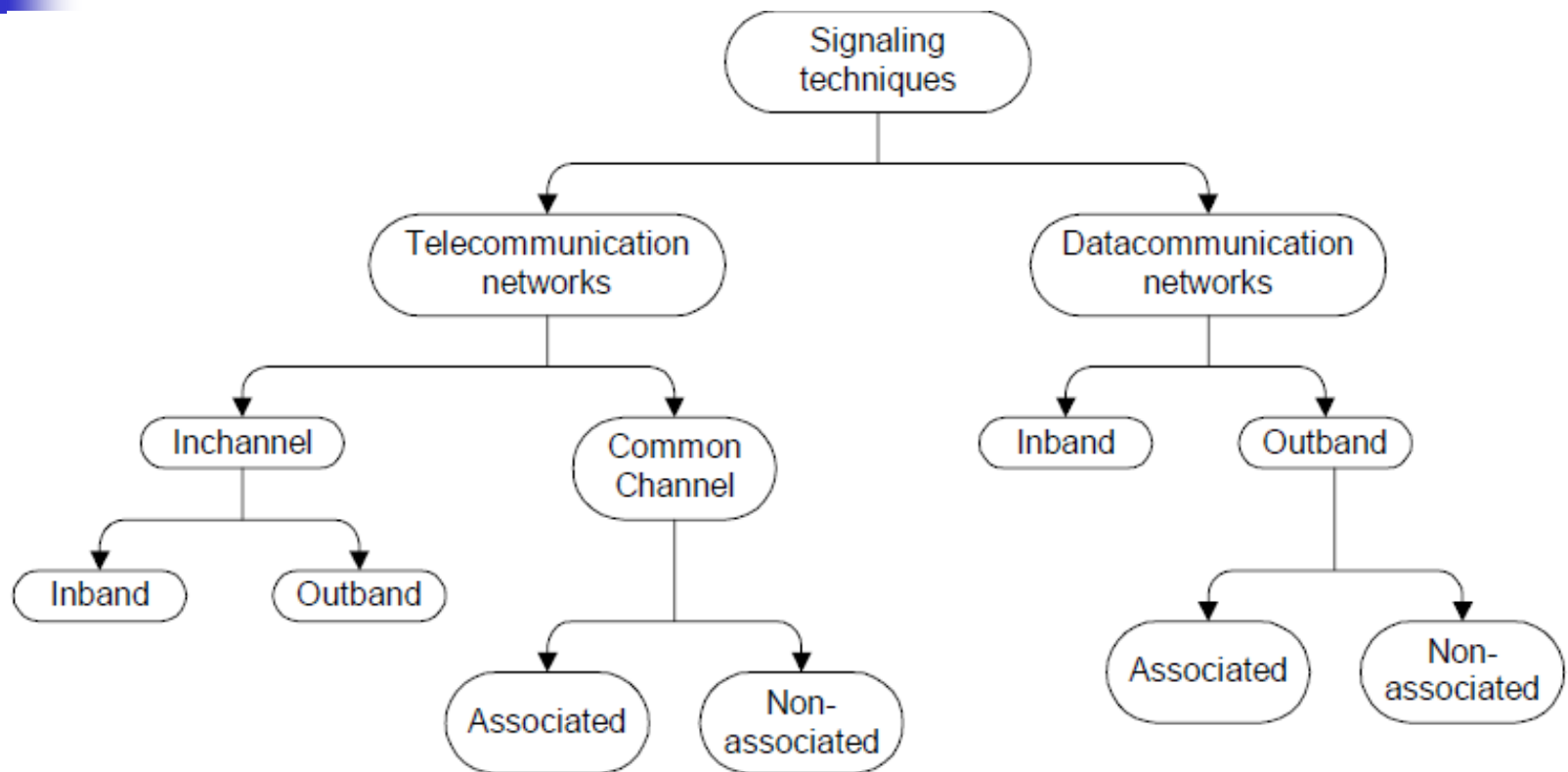
- If Permanent Virtual Circuits (PVCs) are established, generally, no signaling is required.
- For Switched Virtual Circuits (SVCs), signaling takes place using a well-defined signaling protocol.
- For e.g., Q.2931/Q.2971 (signaling protocol for ATM) is much more complicated vis-a-vis Q.933 (signaling protocol for frame relay).



Signaling in Datacom Network

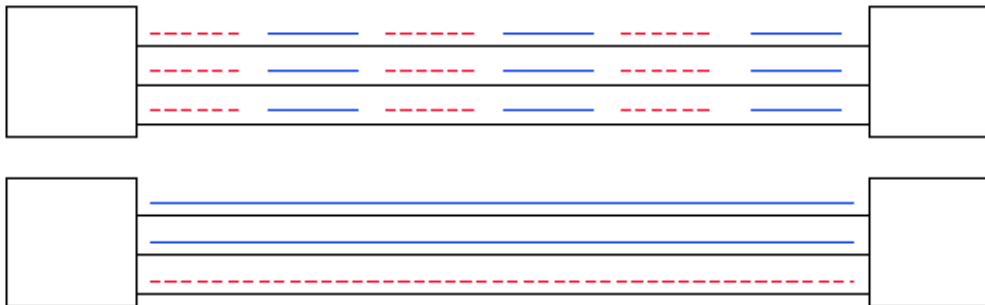
- Datagram networks, generally, do not require signaling. This is because, by definition, a connectionless network does not entail connection setup.
- To provide QoS, some resource reservation and, hence, some form of signaling is required.
- For e.g., the protocols like MPLS and RSVP require some form of signaling message exchange and resource reservation.

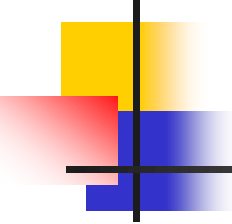
Signaling Channels and Techniques



Signaling Channel

- In-band signaling $\Rightarrow$ Signaling over the same channel as the payload
- Out-of-band signaling $\Rightarrow$ Separate channels for signaling (but maybe the same physical circuits)
- Common Channel Signaling (CCS)
 - Separate circuits for signaling
 - Allows several new functions, such as 800

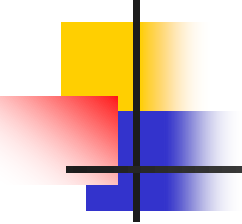




Inband versus Outband Signaling

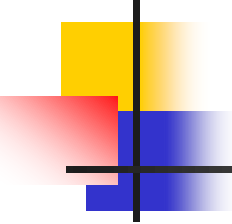
Telecom Network:

- In-band signaling refers to using the same voice frequency band to carry signaling information as that used to carry voice (i.e., 300-3400Hz).
- In contrast, *out-band signaling* refers to using frequencies above the voice band (but below the upper threshold of 4000Hz) to carry signaling information



Datacom Network:

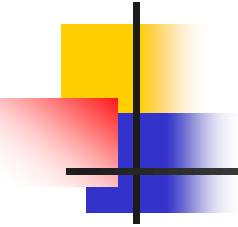
- *Inband signaling* refers to using the same virtual channel to carry signaling information that is used to carry data.
- In contrast, in *Outband Signaling* the signaling information and data are carried on different virtual channels.



Inchannel Signaling versus Common Channel Signaling

Telecom Network:

- In-channel signaling, the same physical channel carries signaling information as well as voice and data.
- In contrast, Common Channel Signaling uses a separate channel for solely carrying signaling information for a number of connections.

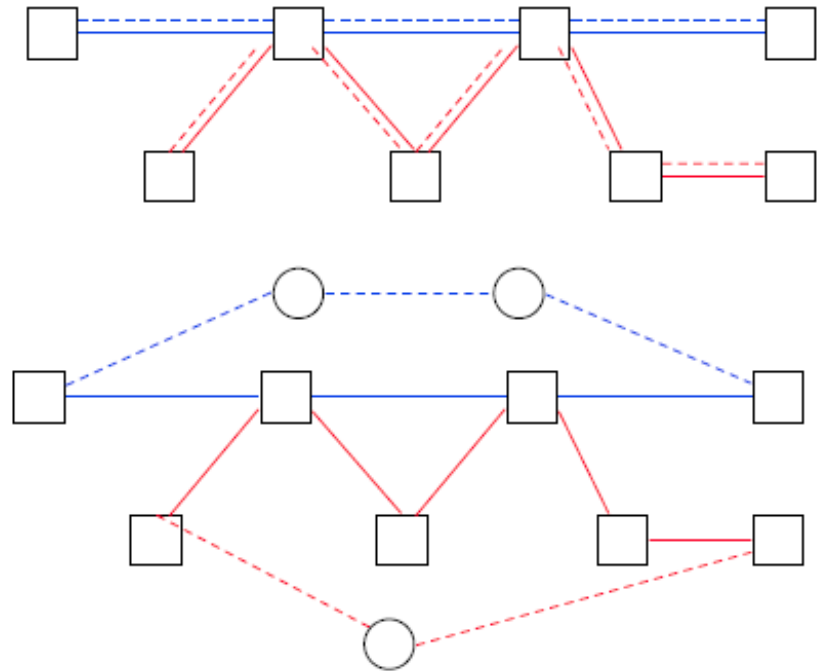


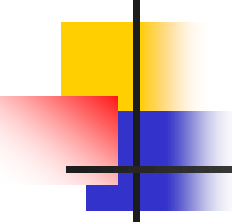
Datacom Network:

- To some extent, in-channel signaling and common channel signaling in telecommunication networks are analogous to in-band signaling and out-band signaling in data communication networks, respectively.

Signaling Modes

- **Associated Modes:**
Common Channel
Signaling (CCS) follows
the same path as the
payload
- **Nonassociated Modes:**
Common Channel
Signaling (CCS) uses a
separate network

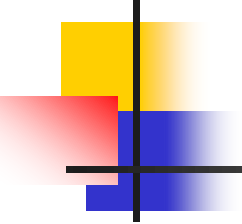




Associated Signaling versus Non-Associated Signaling

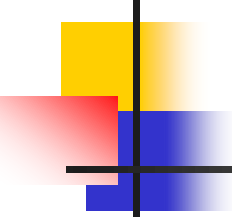
Telecom Network:

- Both these techniques are variants of Common Channel Signaling
- In Associated signaling, the signaling channels and the data paths pass through the same network elements.
- In Non-associated signaling, there is no correspondence between signaling channels and data paths.



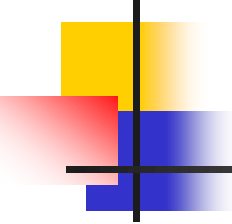
Datacom Network (for ATM):

- In Channel associated signaling, all the signaling messages for each VPI is exchanged on VCI=5 of that virtual path
- In Channel non-associated signaling, all the signaling messages of all the virtual paths are exchanged on VPI=0 and VCI=5.



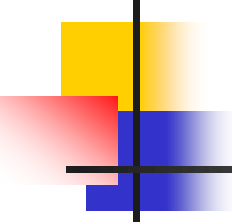
MetaSignaling

- Another technique metasignaling finds mention in various signaling standards
- Metasignaling refers to the process of establishing signaling channels using signaling procedures.
- Used to setup signaling channels.
- All meta-signaling messages are one cell long and have $VPI/VCI = 0/1$



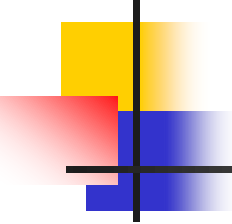
Cont.

- It Sets up 3 types of signaling channels:
 - Point-to-point
 - General broadcast
 - Selective broadcast
- Procedures to:
 - Set up new signaling channels
 - Release channels
 - Verify channels



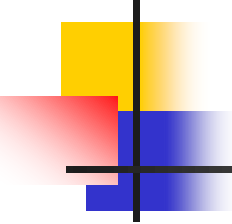
Signaling Issues

- Acknowledgements
- Timer protection
- Parameter negotiation
- Call/Connection identification
- Finite state machine modelling
- Message encoding and decoding (TLV format)

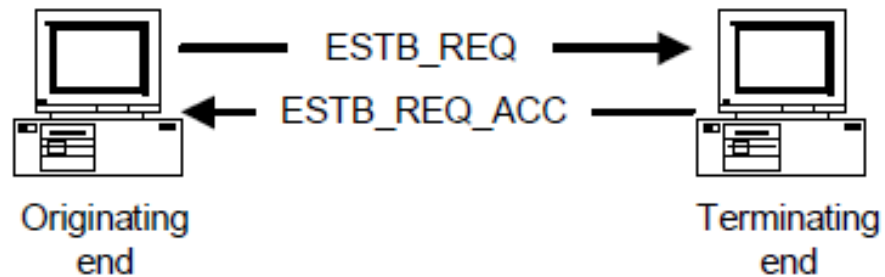


Acknowledgements

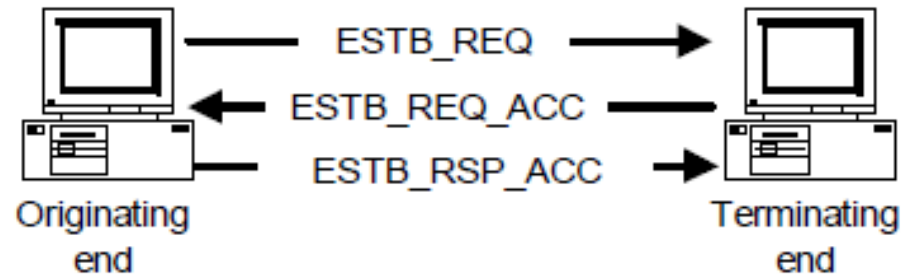
- Required due to unreliable nature of transmission media.
- The classical two-army problem suggests that no scheme can provide full-proof acknowledgement for an unreliable media.
- However, 2 or 3 handshakes is typically sufficient for a normal case.



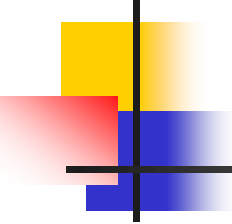
Cont.



Two Way Handshake

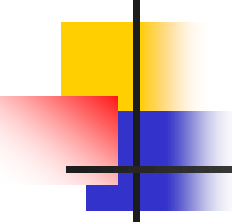


Three Way Handshake



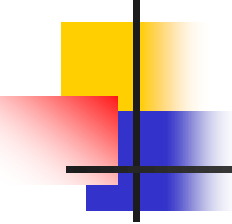
Timer Protection

- Timers are used to avoid inordinate delays in case the signaling messages get lost or corrupted.
- Timer is started after message transmission.
- In case message is lost or discarded, the timer expires and message is retransmitted.
- In case message is lost or discarded, the timer expires and message is retransmitted.



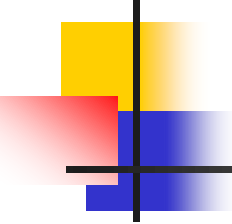
Cont.

- Choosing the correct timeout value is important.
 - If this value is too small, then timers will timeout very frequently.
 - If a very large value is chosen, it may defeat the purpose of keeping timers.
 - Typical value is twice the round-trip propagation time.



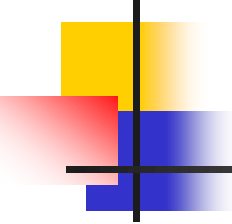
Parameter negotiation

- This entails arriving at a common set of parameters.
- The nature and scope of parameter negotiation depends on the number of handshakes.
- In a two way handshake, negotiation is bare minimal.
- A three way handshake provides more scope for negotiation.



Call/Connection identification

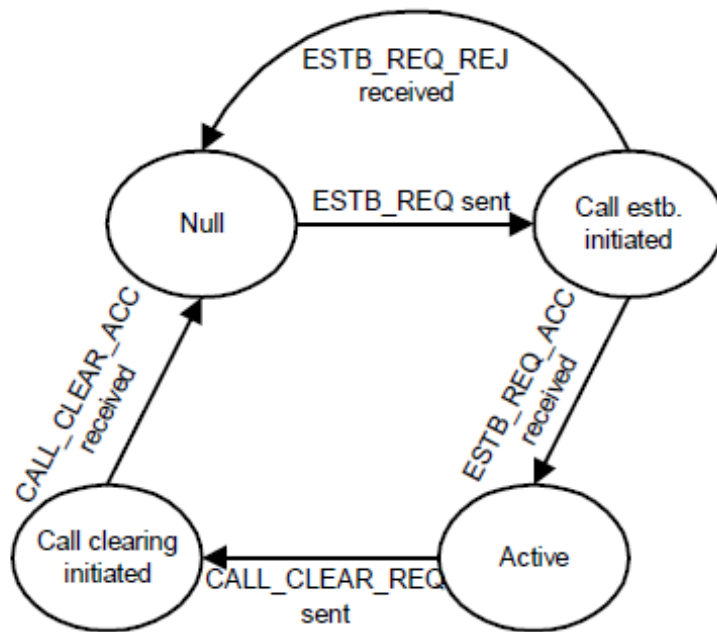
- Consider the following steps:
 - An end-system 'A' sends a connection establishment request to 'B'
 - 'A' sends another request to end-system 'B' for connection establishment
 - 'A' then receives a reply from 'B'. How does 'A' identify to which request has 'B' replied to?
- The solution is to generate a uniq_num and accompany it with every message



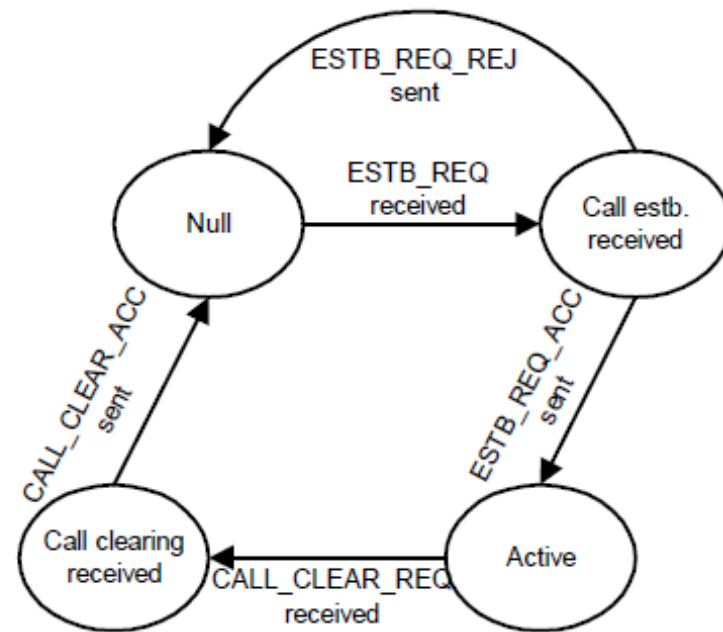
Finite State Machine (FSM)

- A Call goes through three phases
 - Call establishment
 - Data transfer
 - Call releasing
- The FSM accepts messages only if the message is permitted in that state.
- The state change happens when 1) a message is received from peer, 2) Timer expires and 3) User Request.

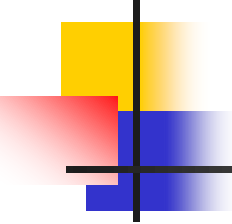
Finite State Machine (FSM) (2)



(a)



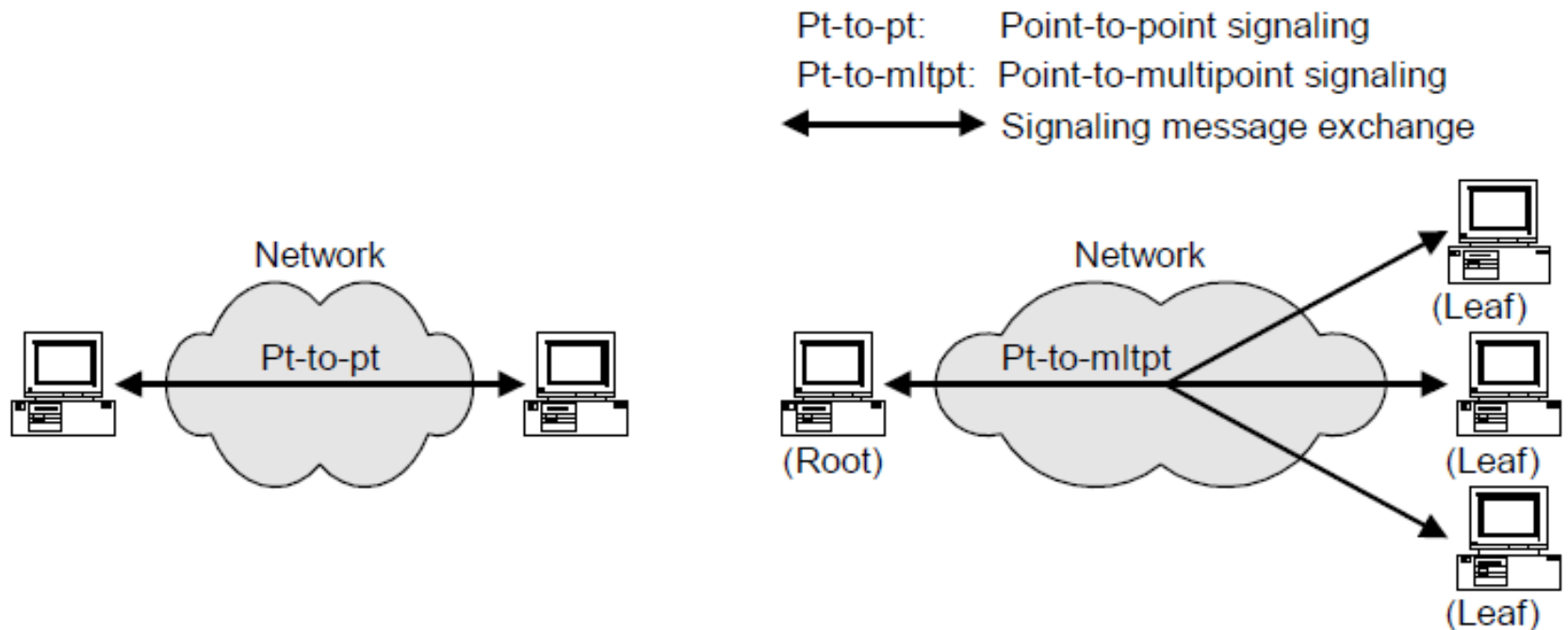
(b)

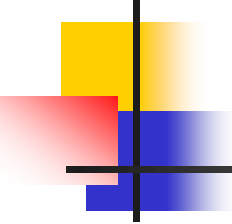


Encoding/Decoding

- Generally, messages are encoded in Type-Length-Value (TLV) format.
 - Type: Identifies the type
 - Length: Length of message (total length or length excluding the header)
 - Value: The actual contents
- Information blocks within the message may also be encoded in TLV format

Signaling Models

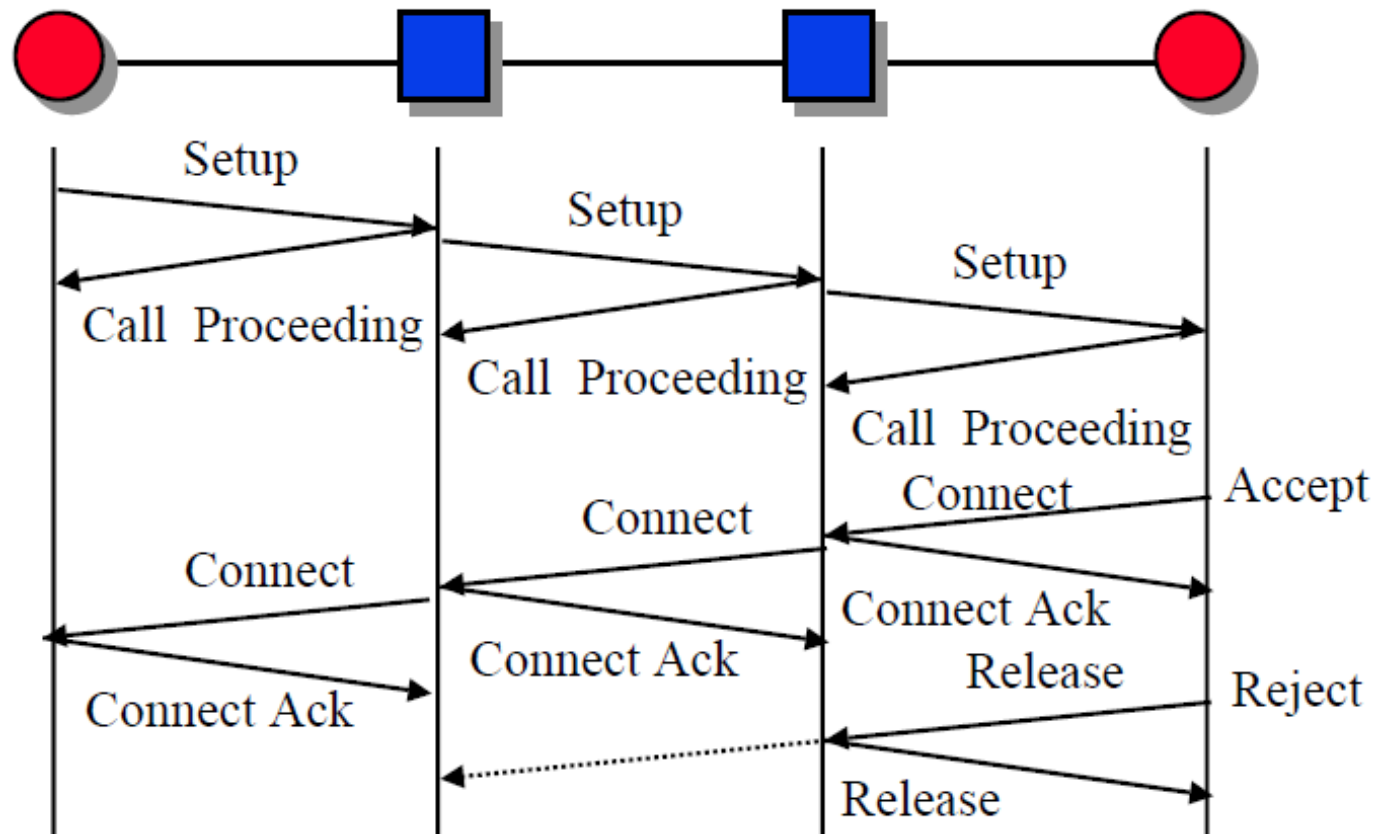




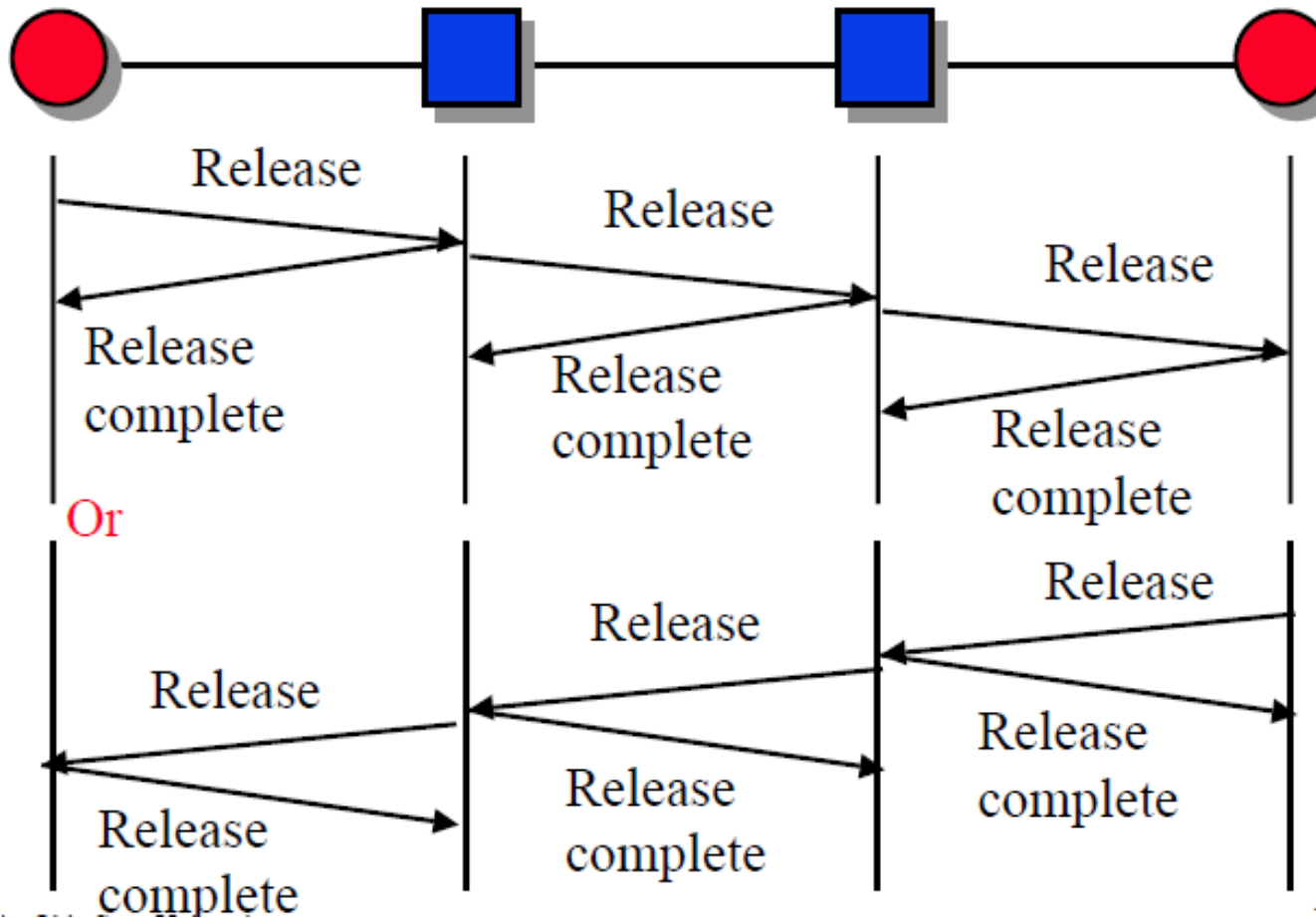
Point-to-Point Signaling

- Point-to-Point signaling model is used to establish and release connections between two end-points.
- This is the most common model of signaling.
- Not only is this model popular, it is also very simple to implement.

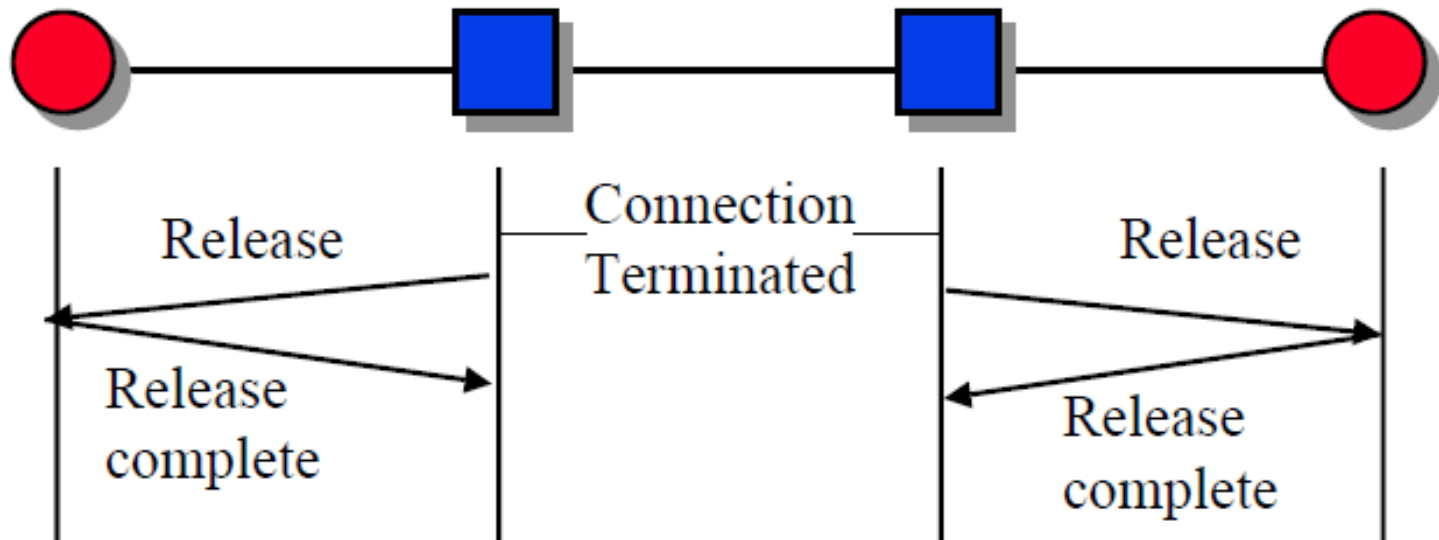
Connection Setup

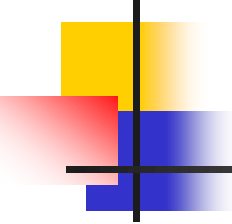


Connection Release



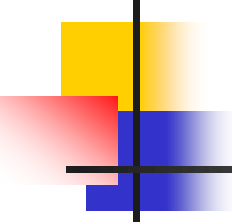
Connection Release (cont)





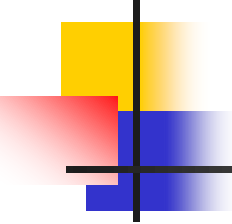
Point-to-MultiPoint (PMP) Signaling

- Point-to-MultiPoint signaling model is used to establish and release connections between a root and multiple end-points.
- This form of signaling is mainly used for multicasting or broadcasting applications (e.g., distant learning).



Starting a PMP call

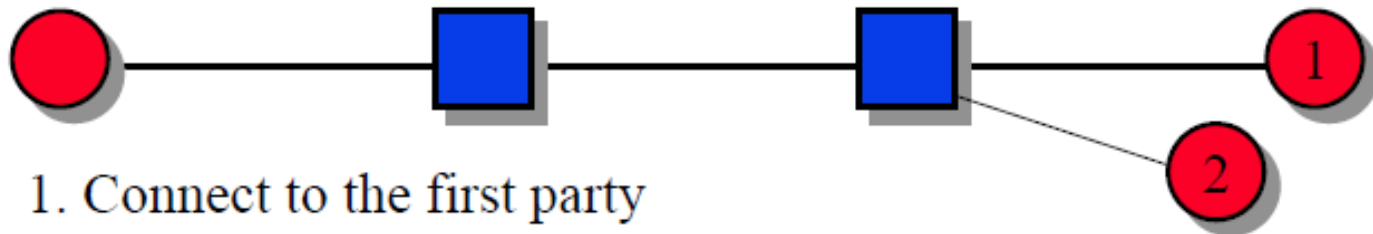
- A PMP call is generally started by the root.
 - The root may take this step voluntarily,
 - Or, it may do the same after receiving an explicit request from a leaf.
 - The leaf can send the connection establishment request to the root through signaling channel or through other means.
- The first connection is established following point-to-point procedures.



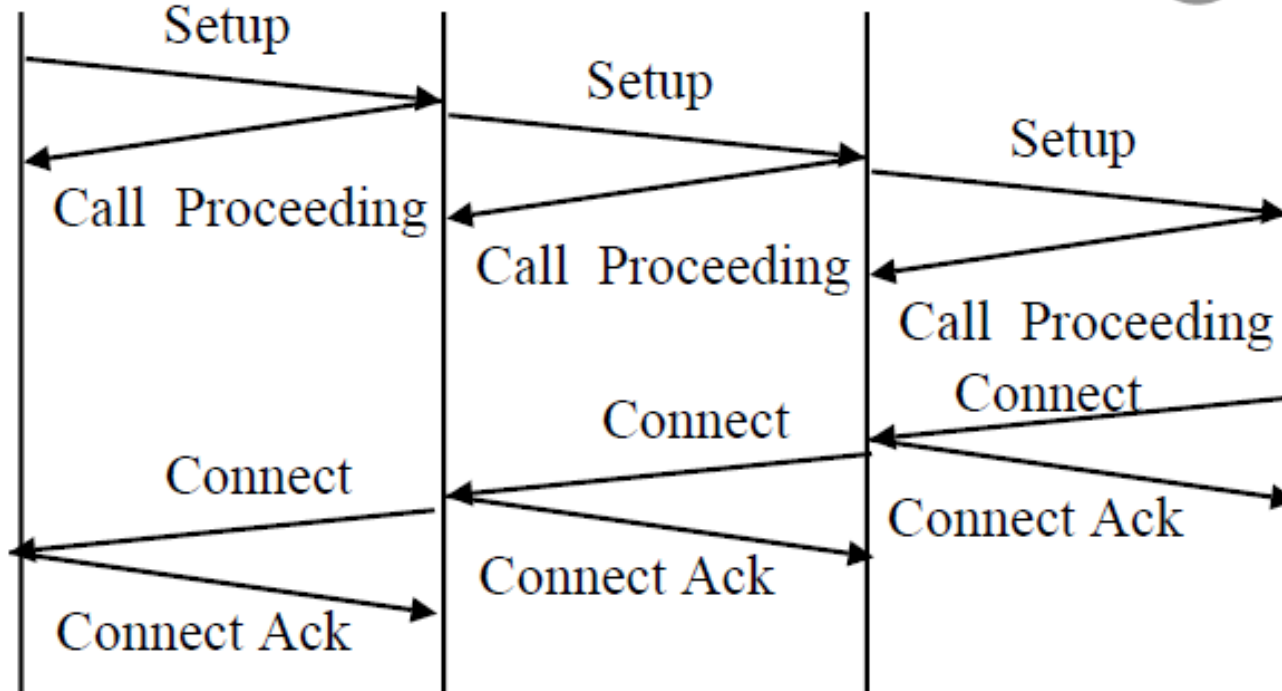
Adding parties to a PMP call

- After a PMP call is established, parties are added by the root.
- The root is informed either through a signaling message, or through some other means.
- Subsequent parties have no say in determining the parameters of the connection, as it has already been fixed

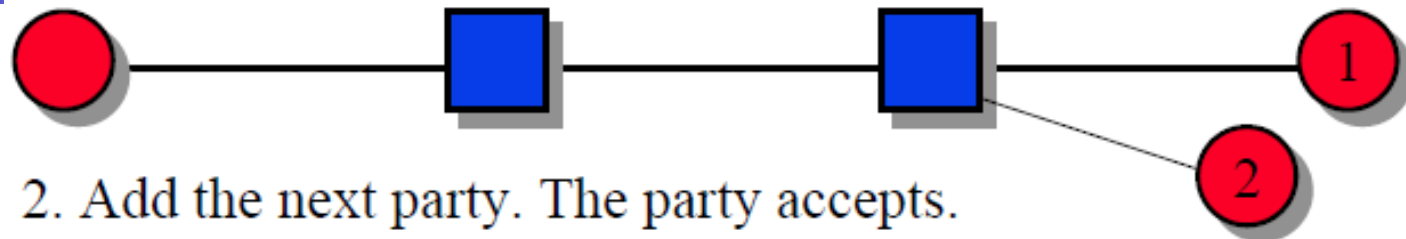
Multipoint Connection Setup



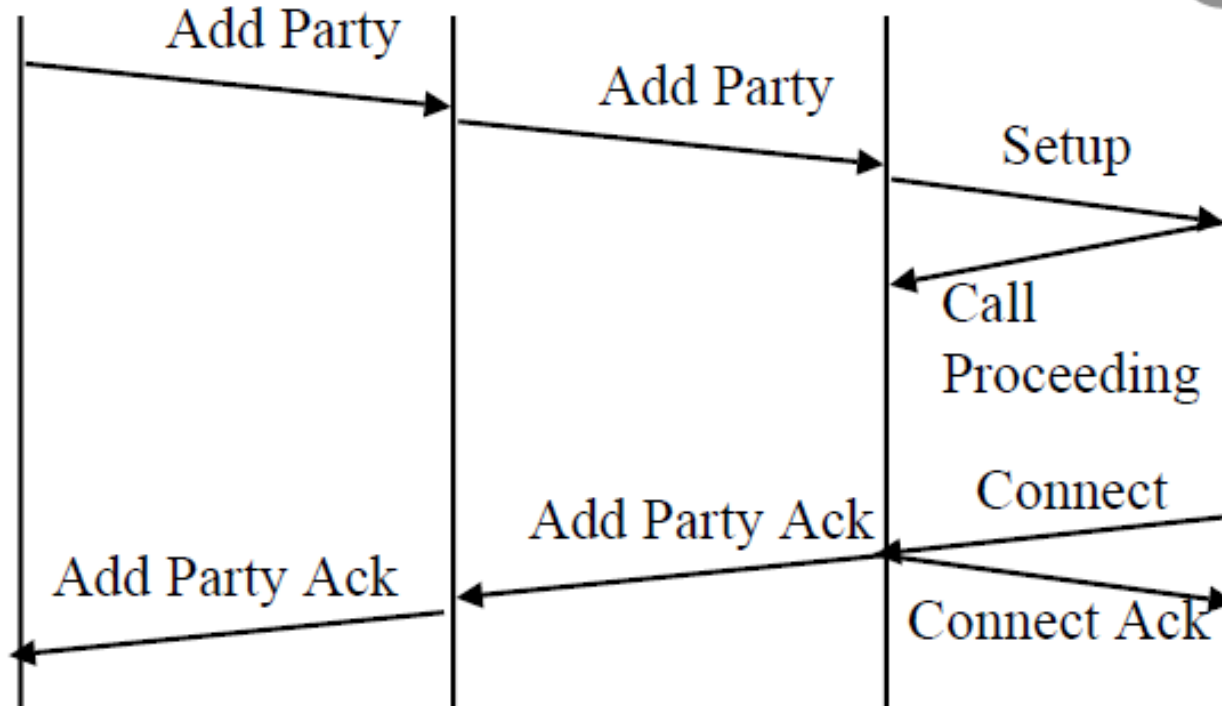
1. Connect to the first party



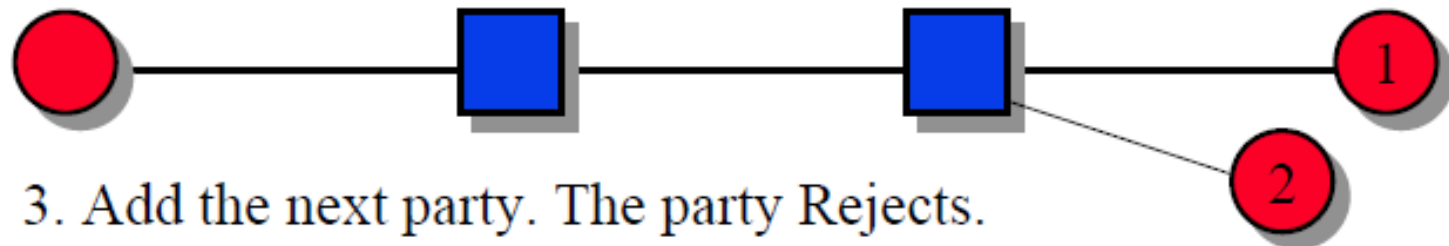
Multipoint Connection Setup (cont)



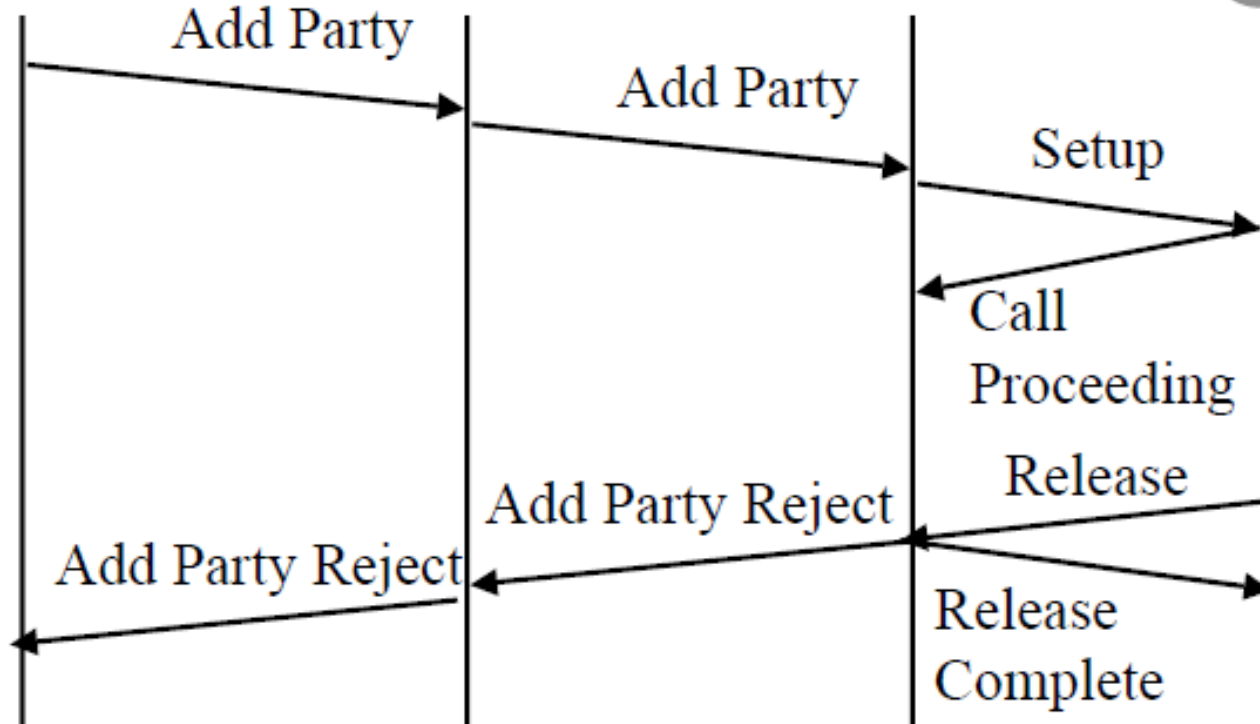
2. Add the next party. The party accepts.

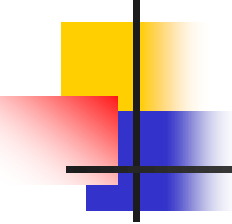


Multipoint Connection Setup (cont)



3. Add the next party. The party Rejects.

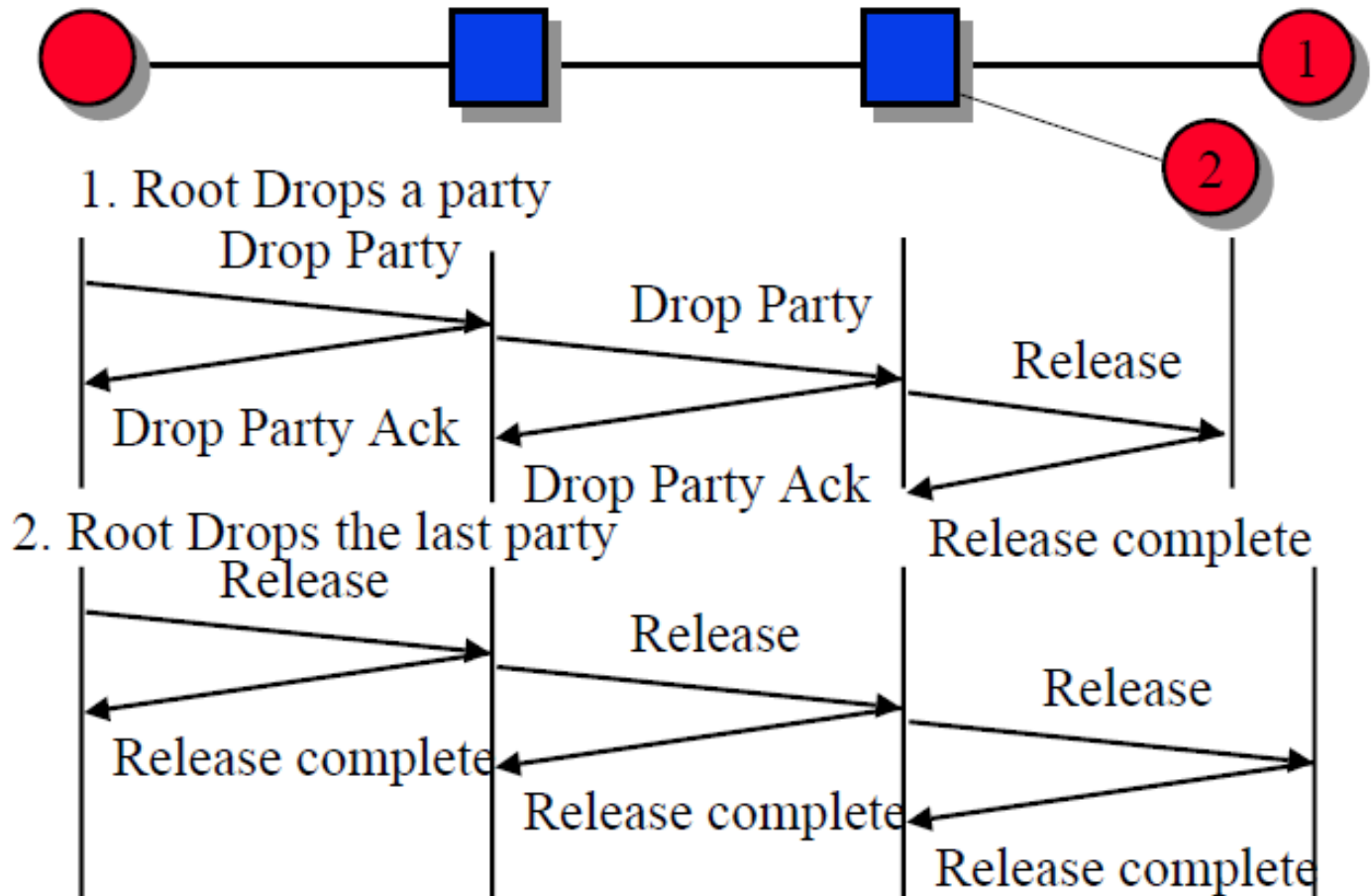




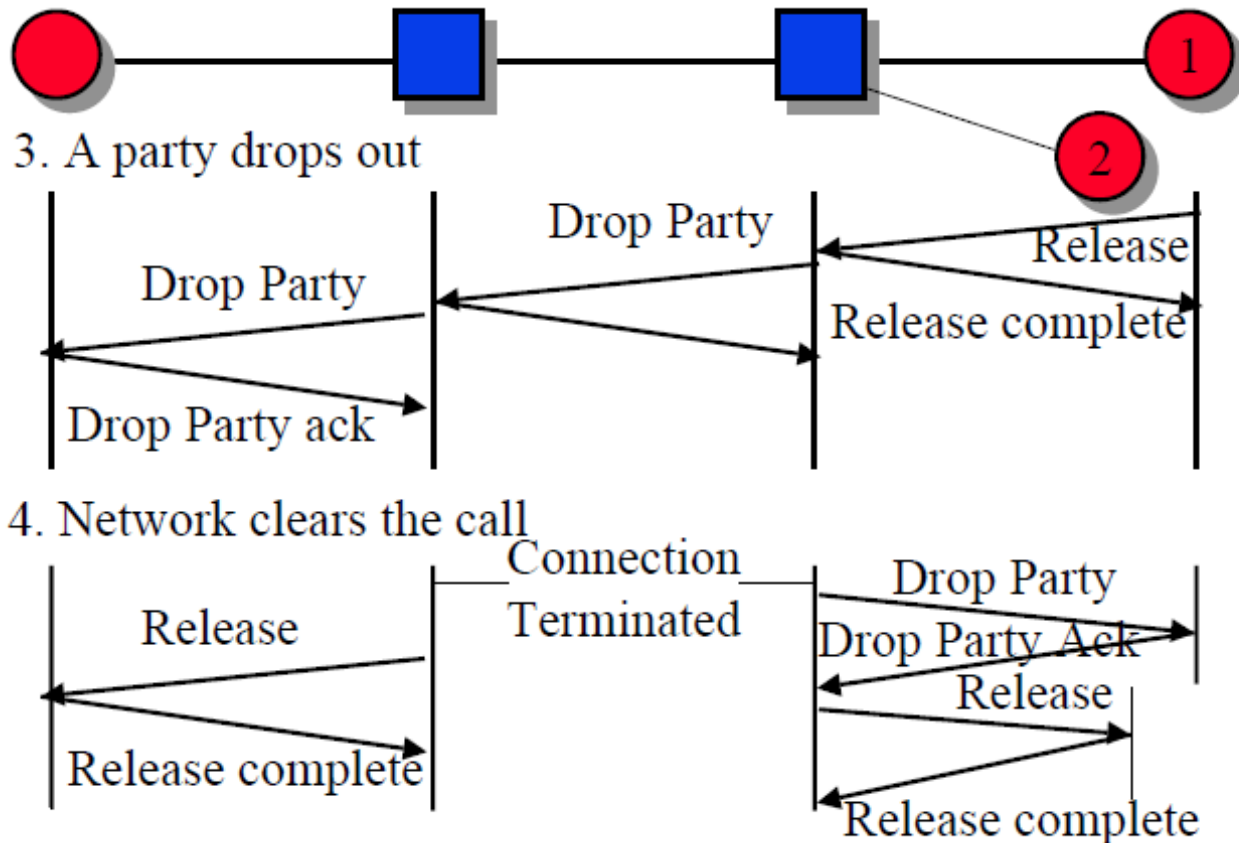
Dropping parties and releasing a PMP call

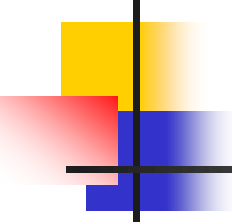
- A leaf of a point-to-multipoint call can drop itself out of the connection by sending a message to the root.
- It is mandatory for the root to entertain this request, and drop that particular party.
- If the root drops itself out of the connection, the whole connection is cleared.

Multipoint Connection Release



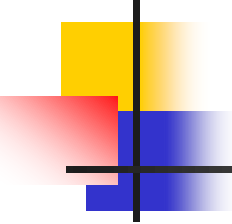
Multipoint Connection Release (cont)





Nature of a PMP call

- By definition, a PMP call is one in which the data flows from the root to the leaves (i.e. unidirectional in nature).
- Theoretically, nothing precludes bidirectional data-flows in PMP calls.
- However, if leaves are allowed to send data to the root, there is a multipoint-to point connection along with the point-to multipoint call.



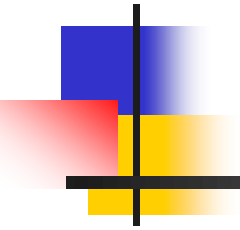
Adv/Disadv of a PMP call

- Advantages

- Saving of network resources like bandwidth. The % saving depends upon the breadth and the depth of the PMP tree.

- Disadvantages

- PMP connections are difficult to establish, manage, and release.
- It is not easy for the leaves to indicate the root to start a connection.
- The unidirectional nature



Thank You!